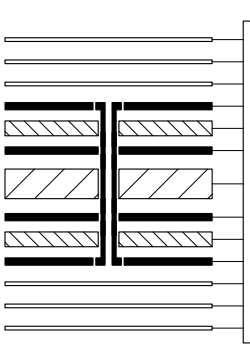
PHASETRACKER Fabrication Document

Layer Stack Legend



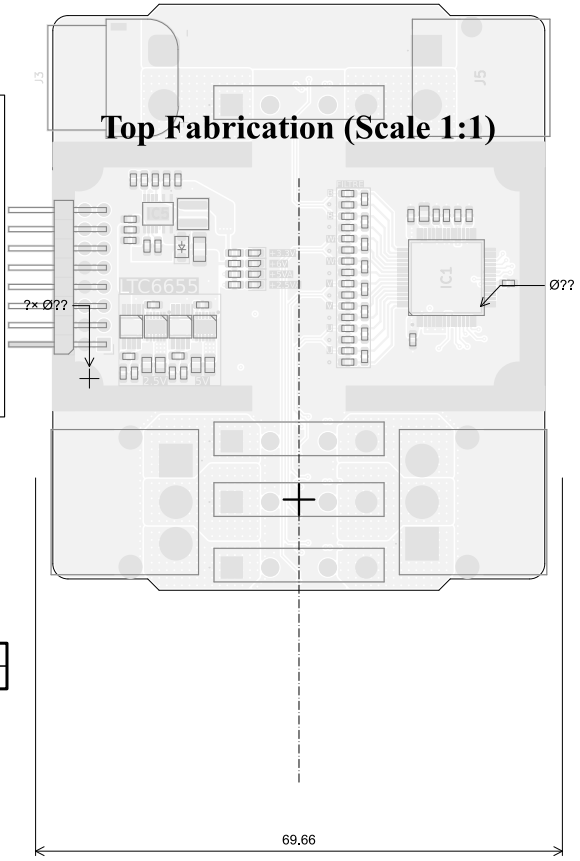
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**L2 (GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Core		1.21mm	FR4	Dielectric	
Copper	**L3 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**L6 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR WHITE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING BLACK NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY OPEN THRUST LAB.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	65.050 × 77.500 mm
BOARD THICKNESS	1.660 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	N/A mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

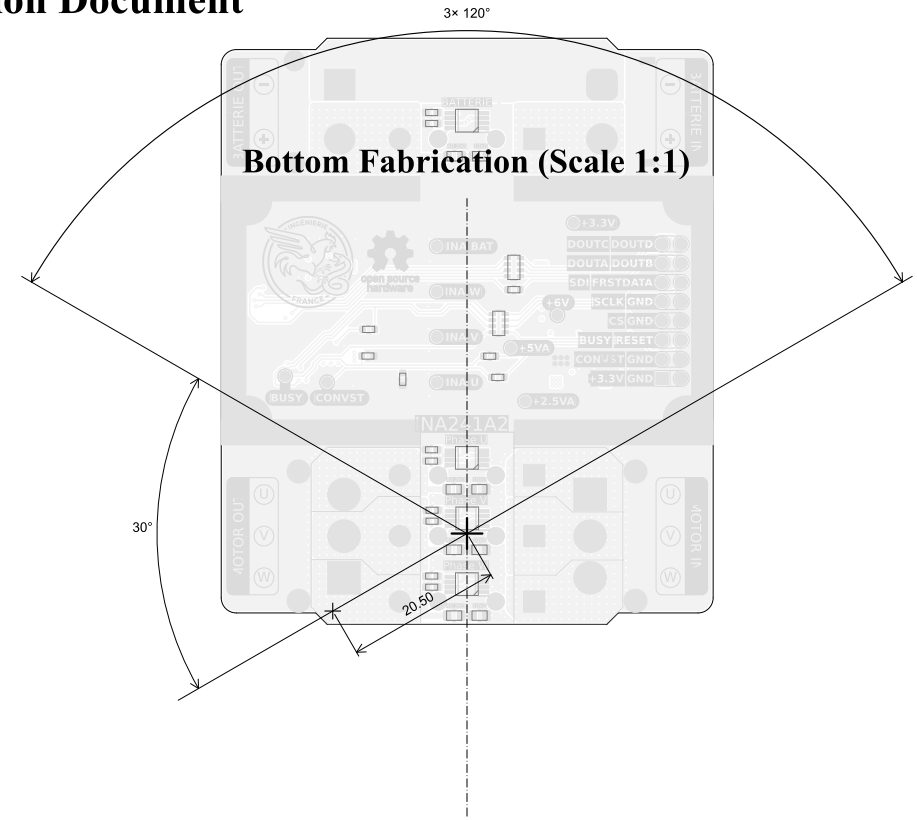
12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: OPEN THRUST LAB		Variant: PRELIMINARY	Git Hash: 4cc6142
	Board Name: **PHASETRACKER**		Project Name: **OPEN_PHASE**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **1** of **9**

PHASETRACKER Fabrication Document

Bottom Fabrication (Scale 1:1)



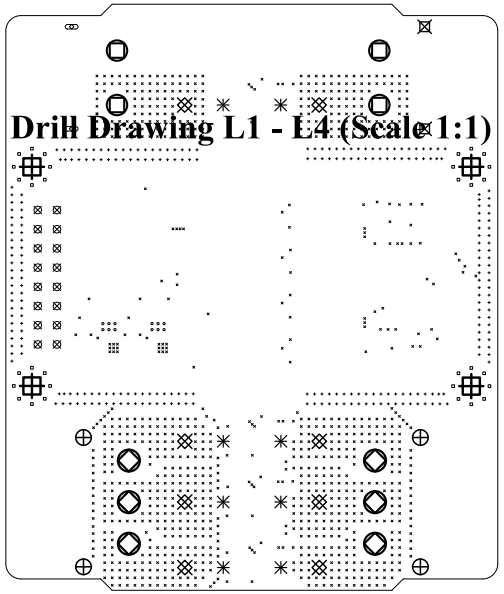
All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: OPEN THRUST LAB		Variant: PRELIMINARY	Git Hash: 4cc6142
		Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 9

PHASETRACKER Fabrication Document

Drill Table

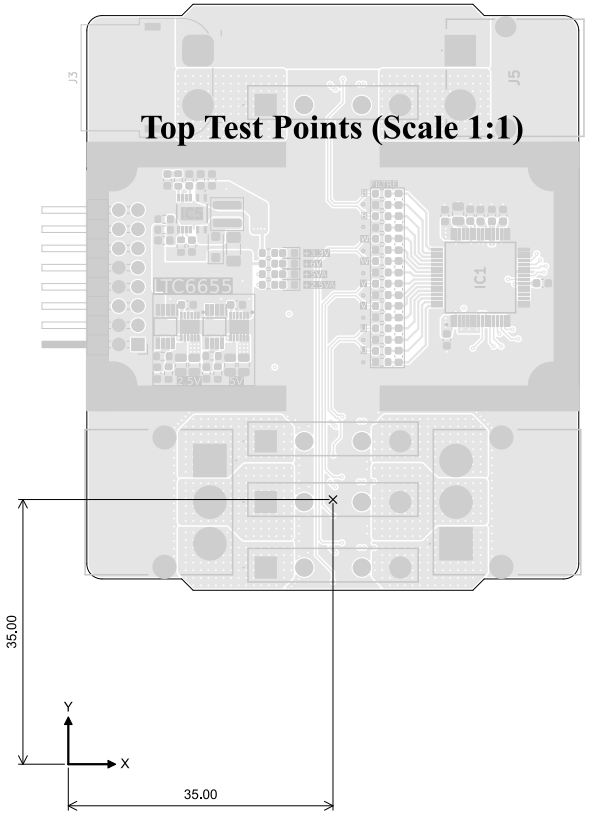
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	937	0.25mm (9.84mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
○	12	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
+	253	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
□	32	0.50mm (19.69mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
◇	2	0.60mm (23.62mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	16	1.00mm (39.37mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
✳	8	1.56mm (61.42mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	2	1.80mm (70.87mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	8	1.90mm (74.80mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊕	4	2.06mm (81.10mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
○	4	2.70mm (106.30mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
○	6	2.90mm (114.17mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊕	4	3.20mm (125.98mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
	Total 1288					



	Comments:	Company: OPEN THRUST LAB		Variant: PRELIMINARY	Git Hash: 4cc6142
		Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 9

PHASETRACKER Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP1		-64.00	106.00
TP2		-64.00	106.00
TP7		-64.00	106.00
TP8		-64.00	106.00
TP13		-64.00	106.00
TP14		-64.00	106.00

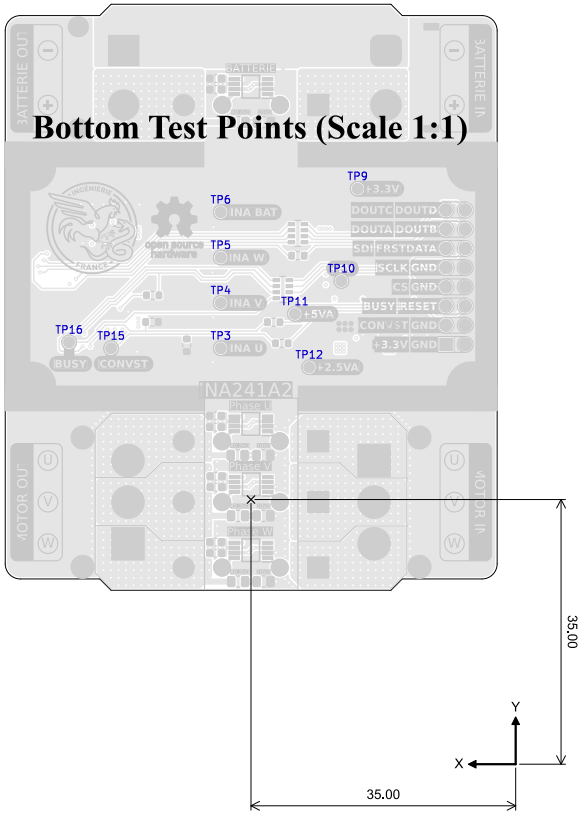
Ref.	Net	X [mm]	Y [mm]

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: OPEN THRUST LAB		Variant: PRELIMINARY	Git Hash: 4cc6142
		Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 9

PHASETRACKER Fabrication Document

Bottom Test Points (Scale 1:1)

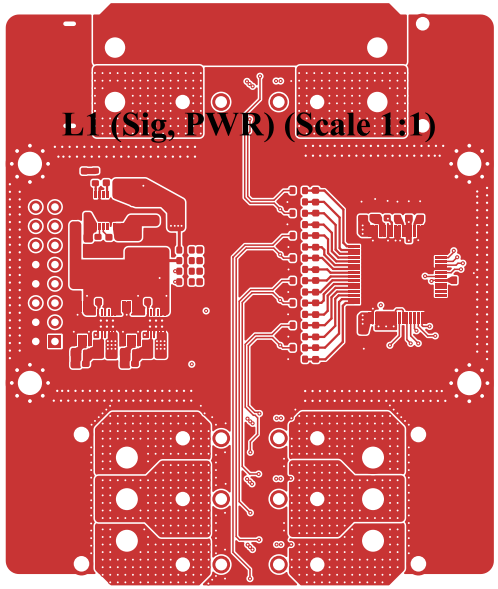


Ref.	Net	X [mm]	Y [mm]
TP3	INA_U	88.50	36.00
TP4	INA_V	88.50	42.00
TP5	INA_W	88.50	48.00
TP6	INA_BAT	88.50	54.00
TP9	+3.3V	70.40	57.10
TP10	+6V	72.55	44.90
TP11	+5VA	78.70	40.55
TP12	+2.5VA	76.80	33.50
TP15	CONVST	103.00	36.00
TP16	BUSY	108.55	36.80

All dimensions are in millimeters unless otherwise specified.

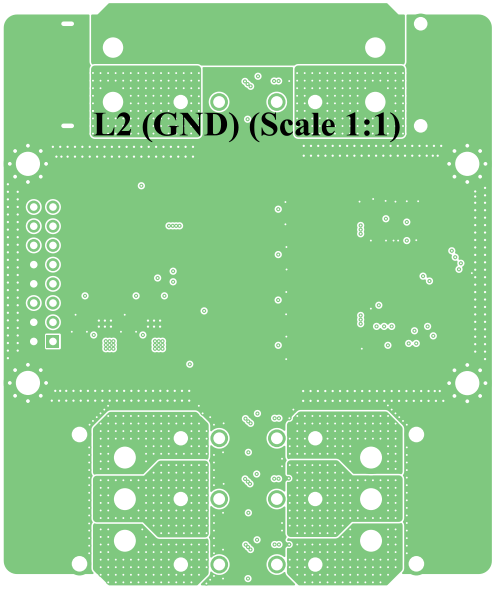
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	Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 9

PHASETRACKER Fabrication Document



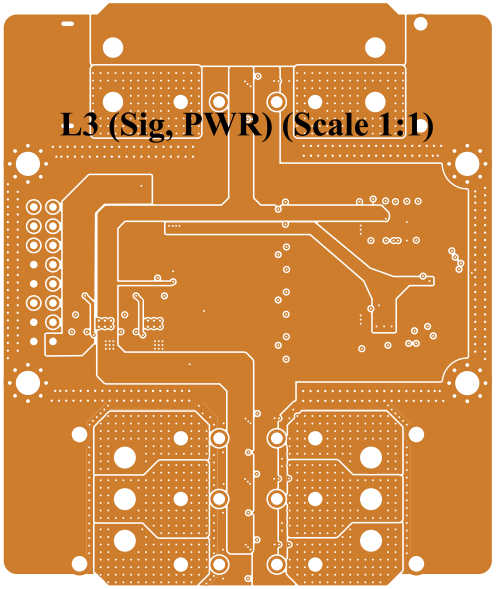
	Comments:	Company: OPEN THRUST LAB		Variant: PRELIMINARY	Git Hash: 4cc6142
		Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 9

PHASETRACKER Fabrication Document



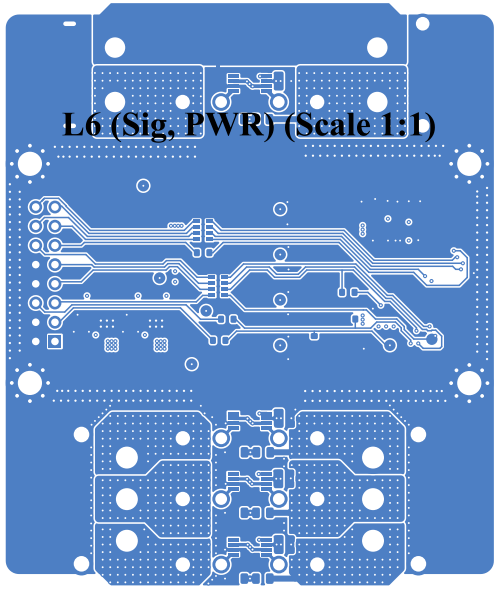
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		Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
	Sheet Title: L2 (GND) (Scale 1:1)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 9

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	Comments:	Company: OPEN THRUST LAB		Variant: PRELIMINARY	Git Hash: 4cc6142
		Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
	Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 8 of 9

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	Comments:	Company: OPEN THRUST LAB		Variant: PRELIMINARY	Git Hash: 4cc6142
		Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
	Sheet Title: L6 (Sig, PWR) (Scale 1:1)	File Name: OPEN_PHASE.kicad_pcb	Designer: SIEBERT DIMITRY	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 9 of 9